(the water's surface becomes concave; Homer feels pressed against the bucket wall; your arms splay outward when you spin around; the rope tied between two spinning rocks becomes taut) would hold true if the experiment were carried out in empty space. This assumption led him to search for something in empty space relative to which the motion could be defined, and the something he came up with was space itself. Mach strongly challenged the key assumption: He argued that what happens in the laboratory is not what would happen in completely empty space.

Mach's was the first significant challenge to Newton's work in more than two centuries, and for years it sent shock waves through the physics community (and beyond: in 1909, while living in London, Vladimir Lenin wrote a philosophical pamphlet that, among other things, discussed aspects of Mach's work[11]). But if Mach was right and there was no notion of spinning in an otherwise empty universe—a state of affairs that would eliminate Newton's justification for absolute space—that still leaves the problem of explaining the terrestrial bucket experiment, in which the water certainly does take on a concave shape. Without invoking absolute space—if absolute space is not a something—how would Mach explain the water's shape? The answer emerges from thinking about a simple objection to Mach's reasoning.

Mach, Motion, and the Stars

Imagine a universe that is not completely empty, as Mach envisioned, but, instead, one that has just a handful of stars sprinkled across the sky. If you perform the outer-space-spinning experiment now, the stars—even if they appear as mere pinpricks of light coming from enormous distance — provide a means of gauging your state of motion. If you start to spin, the distant pinpoints of light will appear to circle around you. And since the stars provide a visual reference that allows you to distinguish spinning from not spinning, you would expect to be able to feel it, too. But how can a few distant stars make such a difference, their presence or absence somehow acting as a switch that turns on or off the sensation of spinning (or more generally, the sensation of accelerated motion)? If you can feel spinning motion in a universe with merely a few distant stars, perhaps that means Mach's idea is just wrong—perhaps, as assumed by